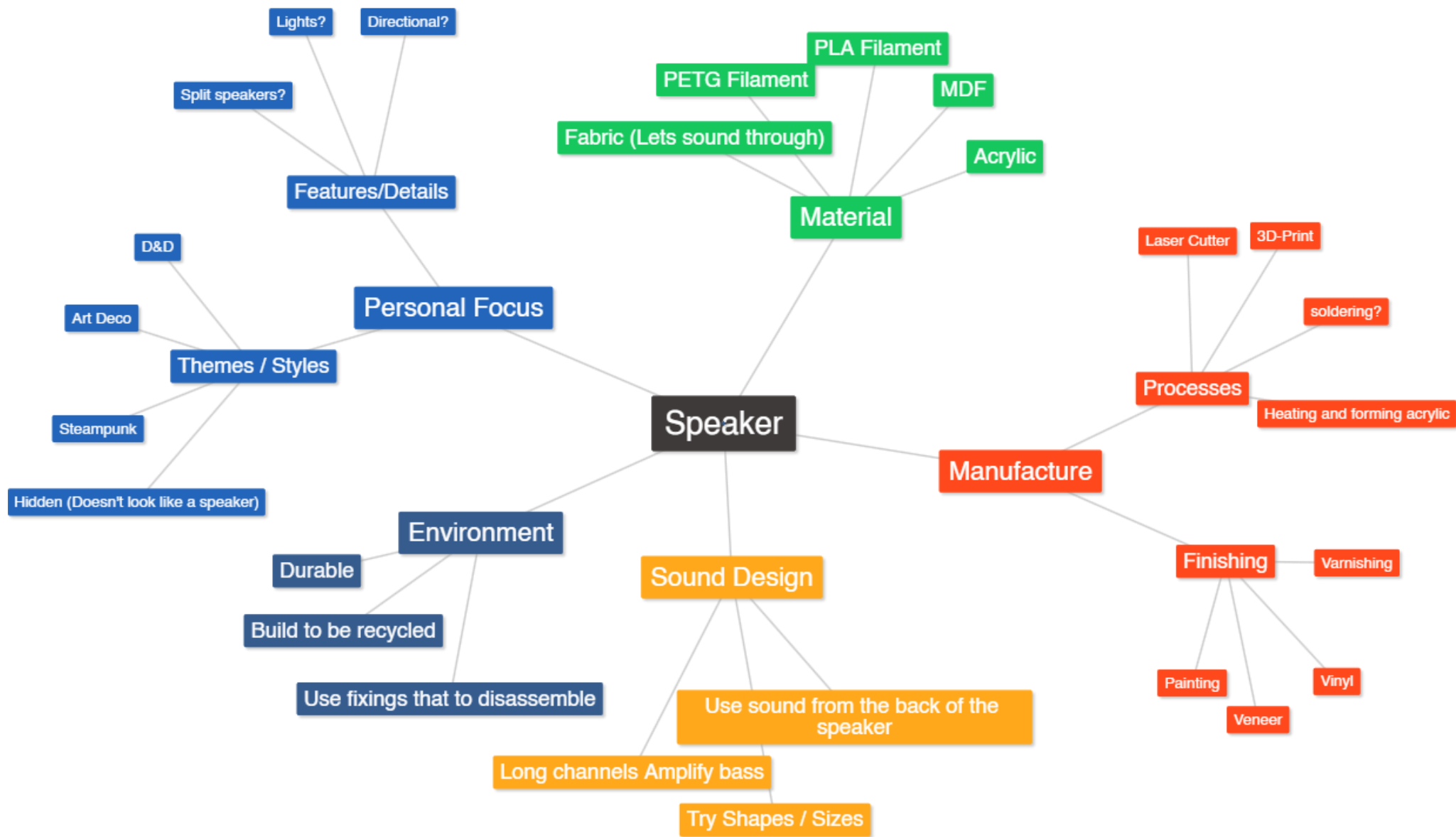




AMPLIFY - Bluetooth Speaker

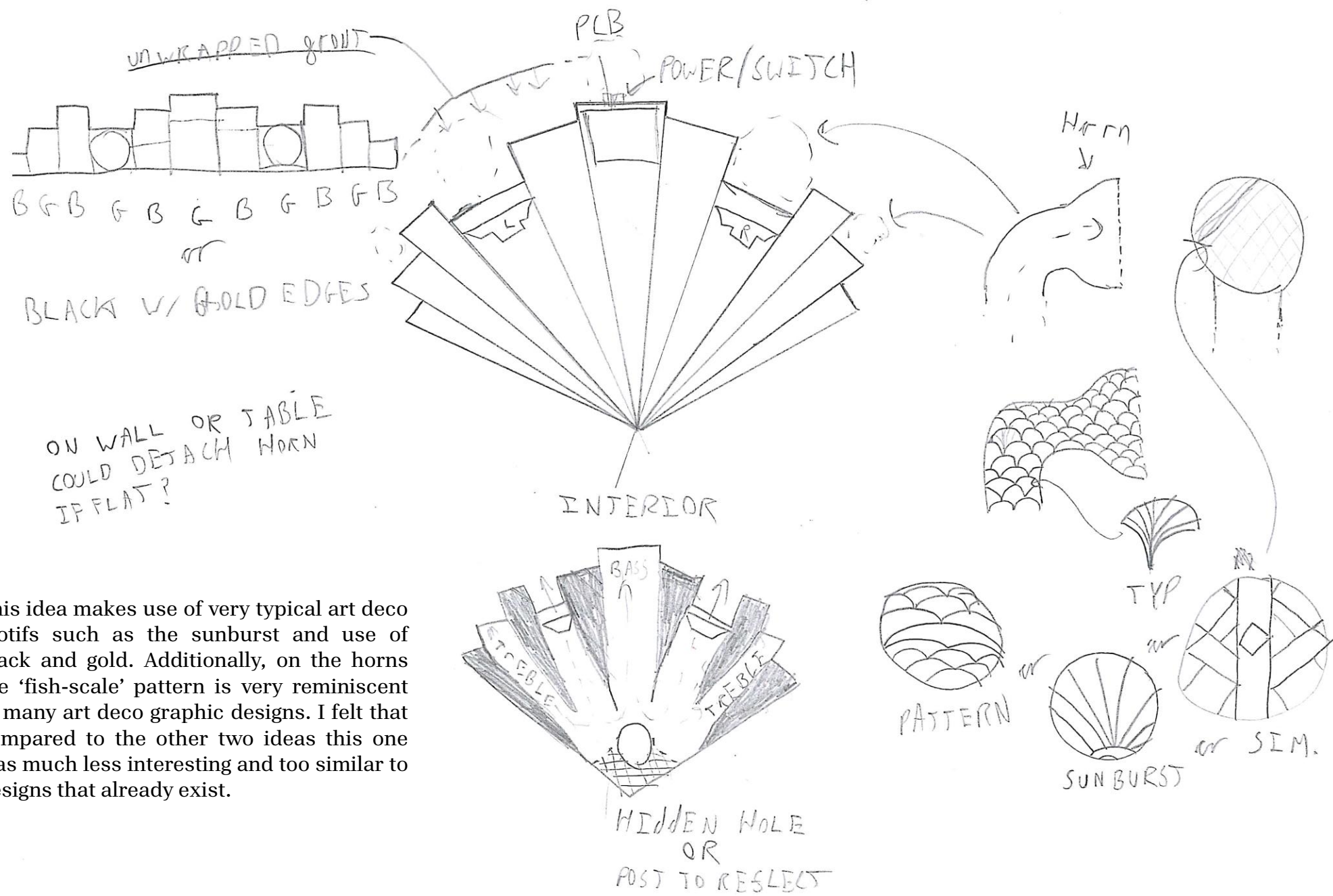


Moodboard

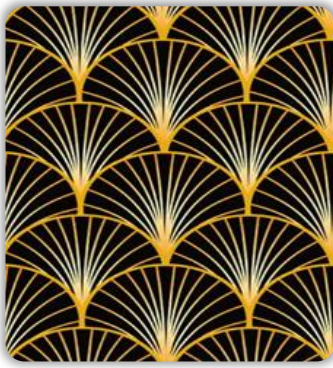
This is a collection of images showing a variety of inspiration based on the personal focus part of my mind map. They show ideas from D&D, including props and dice. They also show steampunk objects and objects made in an art deco style.



Art Deco Idea



A typical Art Deco sunburst



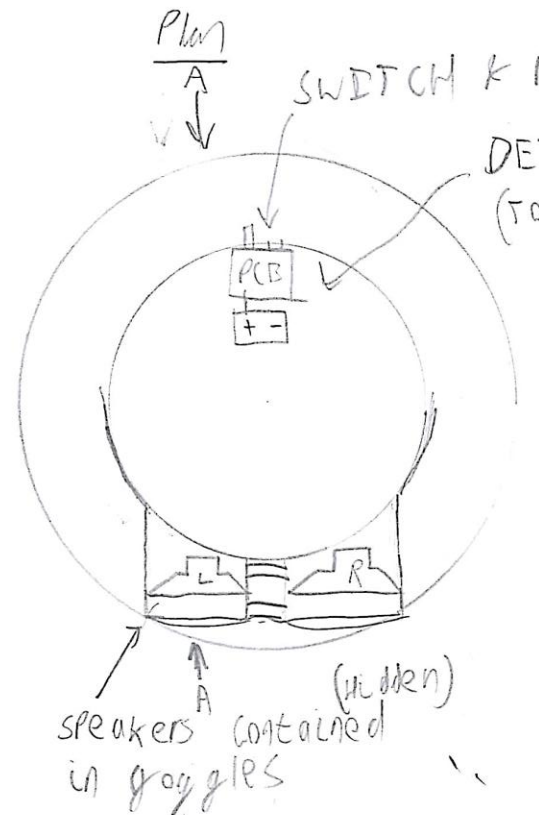
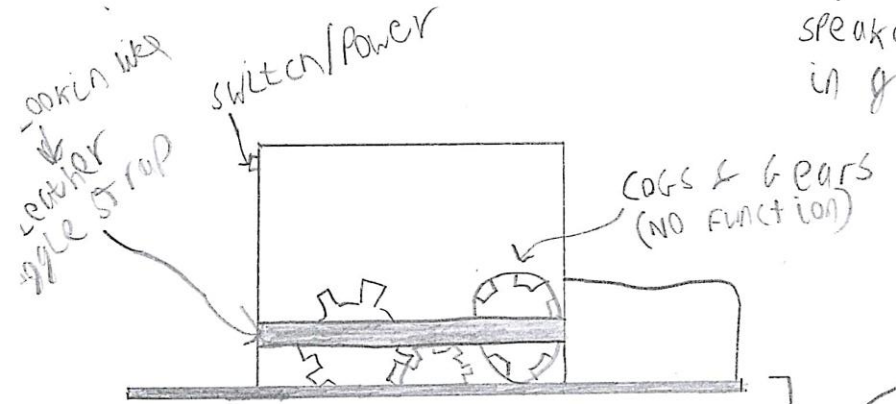
This idea makes use of very typical art deco motifs such as the sunburst and use of black and gold. Additionally, on the horns the 'fish-scale' pattern is very reminiscent of many art deco graphic designs. I felt that compared to the other two ideas this one was much less interesting and too similar to designs that already exist.

Steampunk Top Hat

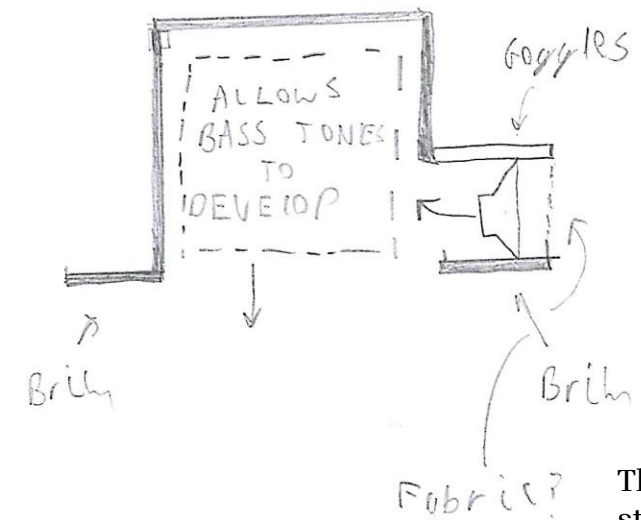


STEAMPUNK goggles

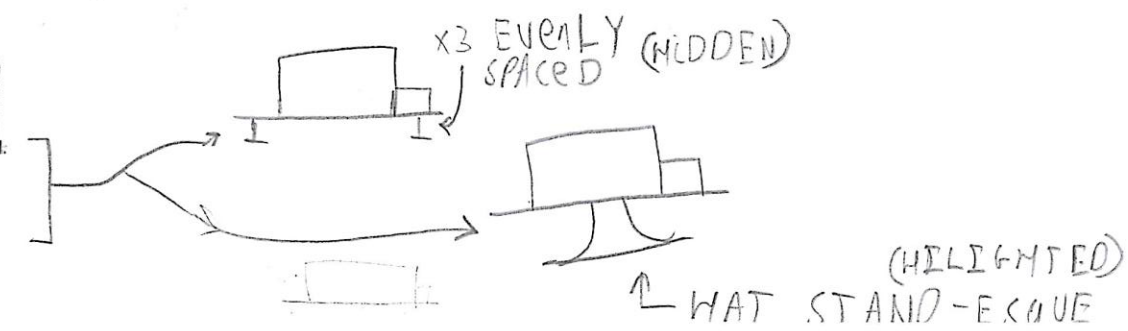
SIDE



CONSTANT USE OF STEAMPUNK ELEMENTS LIKE GEARS (A COPPER/BRONZE)

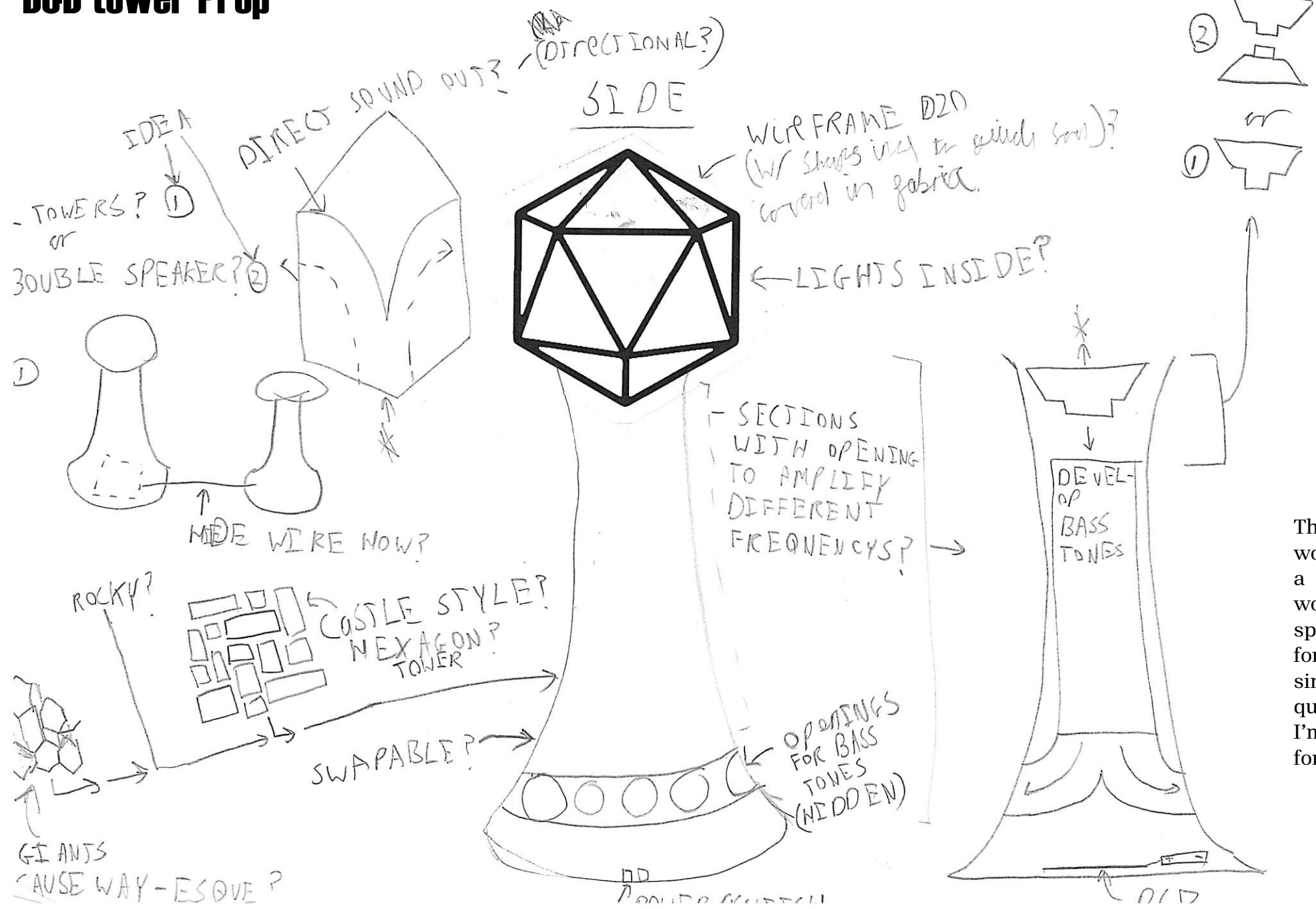


This style of steampunk goggles



This idea makes use of a handful of steampunk motifs, including the gears, goggles and top hat. One fun thing about this idea is that the speakers would be very well hidden and it's not clear that it's a speaker. And it could provide a lot of space inside to develop the sound. It is also a design that could look very aesthetically pleasing, particularly if there were other similarly styled objects around.

D&D tower Prop



Partly Inspired by a D20

This idea is D&D themed and would make an excellent prop for a dnd session particularly as it wouldn't be clear that it's a speaker. There is lots of potential for interesting aesthetic features since the tower would have to be quite large. This is the idea that I'm going to follow up on and use for the rest of the project.

Research

During my research I found that there are various methods to amplify and control sound to improve its quality. By varying the size and shape of the enclosure you can shape the sound to be louder overall, or to have more or less of a certain range of frequencies. However, the calculations required to figure out the dimensions needed are very complex and require a level of understanding that I wouldn't have been able to attain during this project.

Overall, I found that the best way to amplify a speaker and to give clean (not echoey) bass is to use a transmission line enclosure. This is an enclosure which contains a long winding pathway for the sound to travel along. On a very large scale it can look like the image below.



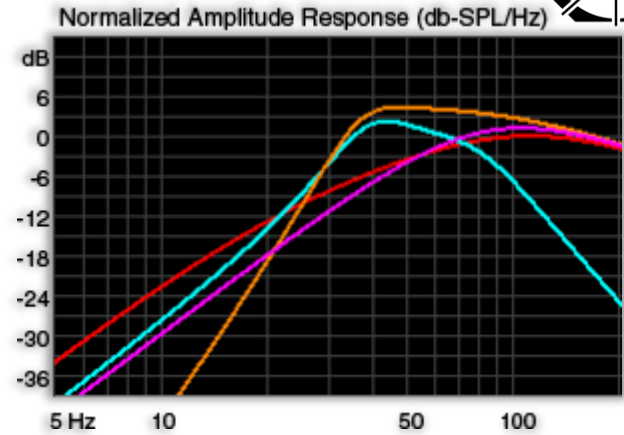
There are also various techniques involving taking multiple speakers and placing them cone-to-cone or magnet-to-cone or magnet-to-magnet, and each variation gives a different sound.

<https://www.kicker.com/how-to-build-a-subwoofer-box>
<http://www.decdun.me.uk/loudspeakers5.html>
<https://www.britannica.com/science/acoustics/Acoustic-problems>
<https://www.diyaudio.com/community/threads/choosing-the-length-frequency-of-a-transmission-line.250098/>

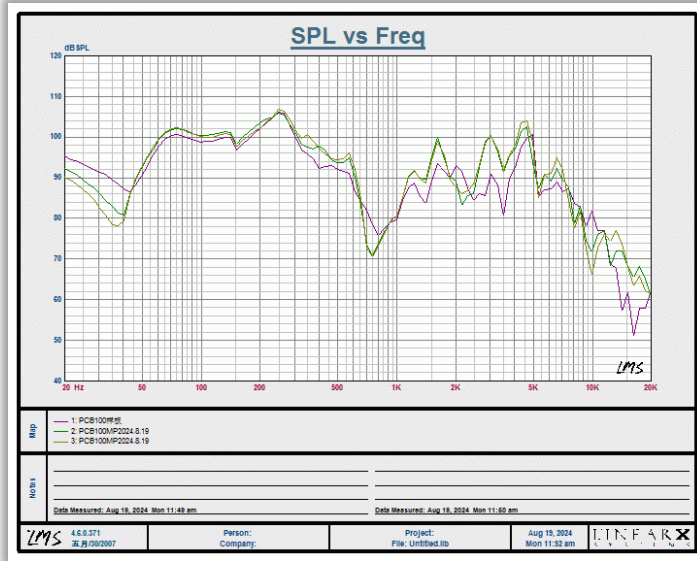
I spoke to someone who has experience working with speakers and sound and they told me some very useful information. Because of how small the drivers are it is important that the enclosure is made of a dense material because you want all the energy to remain as sound and not to be vibrating the enclosure. Additionally, a long internal pathway for sound to travel though amplifies the lower frequencies which is something that will be vital to avoid the speaker sounding tinny. Finally, if there are any places that I don't want sound to travel, I should fill them with foam, again to allow the energy to remain in the sound.



To test the ideas talked about on this slide, I decided to make a cardboard prototype that I can experiment with to gauge what layouts sound best.



Above is a graph of volume against frequency. This shows how different enclosures can affect the sound of a speaker, this shows only low frequencies unlike the graph below which shows the entire range of frequencies for the actual speakers we are going to be working with



Testing



This is the cardboard prototype I made. The speaker can be held on with split pins and it's quite crude, but it allowed me to quickly and efficiently test how different things would sound before making anything more polished. The shape is a pentagon because it was the shape that the tower was originally going to be a pentagon, but this later changed.



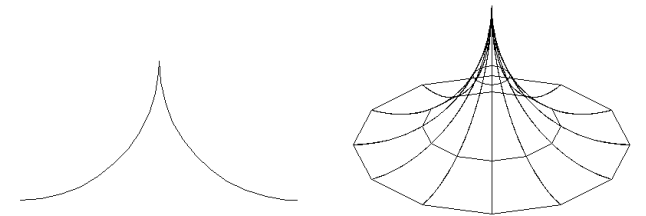
To test this model, I connected both speakers to be PCB and attached one to the model. Then to compare the sound quality I would send audio to only the left or right channel at a time and would listen for the difference. Using the model the sound was much bassier and overall, much better. It was also significantly louder. But it was also introduced a slightly echoey sound so I felt that I could do better. So, I made a second tower and created another piece to insert and to make a transmission line like through the structure. (Right)

The transmission line made it sound even better again; it removed the echo whilst keeping the bass and although was slightly quieter than the other prototype, it was still much louder than the lone speaker. I knew from this point that the internal structure of the tower had to take the form of a transmission line because it just created the best sound.

It was also better raising the bottom of the prism up off the table slightly, I believe this is because it allows the sound to bounce off the table and out around the speaker.



I also experimented with different structures above the speaker and found honestly minor changes. Because of the plan to have a d20 above the top of the tower I felt like it was a good idea to have something hidden inside it to point the sound outwards, but this seems to be quite difficult. What might be a good idea is some kind of cone with curved edges (see below) that directs sound out but that is difficult to manufacture so I will have to return to it later if I need to.

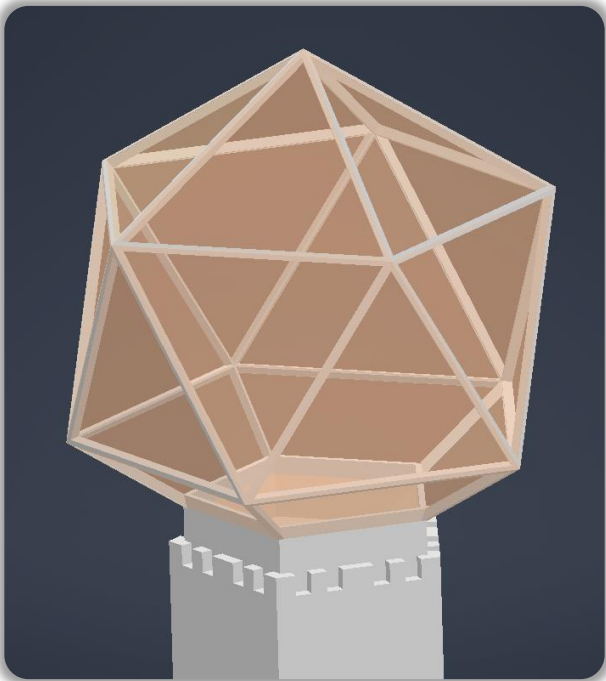


Early CAD tests



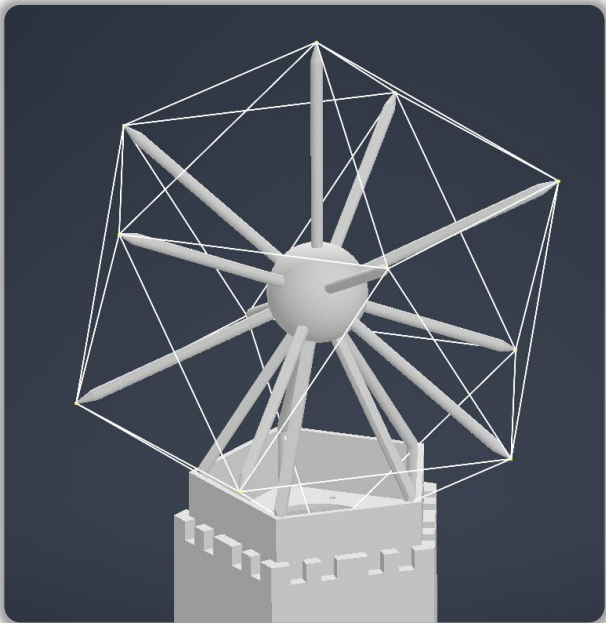
This was the first version of the tower. It was a pentagon so it would match with the 5 edges from the bottom point of the d20. I need some kind of shape to funnel the sound of the bottom and the opening need to be covered with something like fabric. Also, the transmission line hasn't be done yet.

I quite like the idea, but I think it's a little bit dull currently. That said I like the whole castle tower theme so I'm going to stick with that.



Wireframe d20

This is the first version of the d20. It's based on a wireframe of the shape where fabric would then be stretched over it. Manufacturing the wireframe would be very difficult, It would likely have to be 3D printed or laser cut in several sections, and the precision to ensure that it would all fit together would likely be impossible to achieve. For this reason, I'm not sure that this is the best way to make the d20



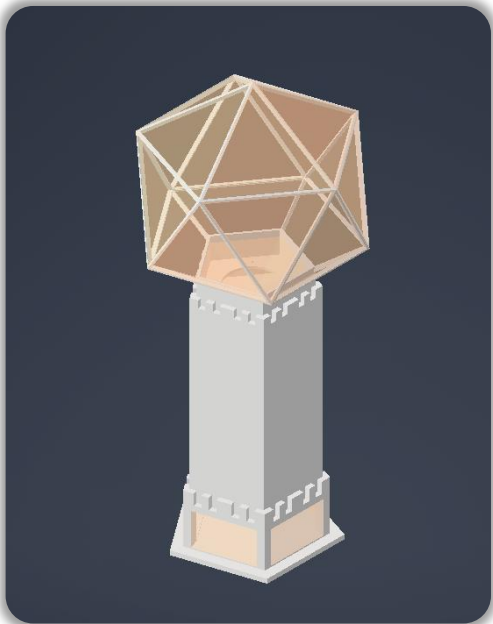
Hedgehog d20

This is an alternate version of the d20. The fabric would be the same, but it would be stretched across dowels that are inserted into a 3D-printed core. This allows for less precision when manufacturing and it would be much easier overall. And the internal structure would bounce sound out and around quite well. However, the fabric would be very difficult to cut to size so I'm still not sure if this is feasible.



No d20?

I think that as much as I love the idea of the d20, with the CAD I've done, and the laser cut test I did for the wireframe variant, it would be very difficult to manufacture. And it makes the tower look quite top heavy. And of course, this could be fixed by making the dice smaller, but it has got me thinking that (especially with the castle theme) I could make the speakers look overall better by creating more interest in the tower and removing the d20.





Exterior

This is the design for the exterior of the speakers. The only component that has a specific place is the speaker which goes right at the top. I think this is a much better design aesthetically than the previous, and that removing the d20 was a good idea because again, it looks much better. I need to create the internal structure, but first there is some other things to consider...

Manufacturing

Before continuing with the design, I need to consider how this is going to be manufactured because not all materials and methods will be able to obtain the same results. This means I need to tailor my manufacturing methods to my needs.

Materials

The materials I have access to are MDF, in 3mm and 6mm sheet form, PMMA (acrylic) in 3mm or 5mm sheet form, and PLA filament for 3D printing.

Density

One part of research showed that I need the material to be as dense as possible to provide the best sound quality, so I have used the information at makeitfrom.com to find the density of the different materials I have access to:

PLA (100% infill)	1.30 g/cm ³
PMMA	1.20 g/cm ³
MDF	0.75 g/cm ³

Ease of use

Both MDF and PMMA can be used with the laser cutter and PLA is used for 3D-printing, so all materials can be quite easy to work with. MDF will take longer to finish if laser cut since it will be burnt but overall, they are quite even in terms of ease of use.

Methods / Techniques

In terms of methods of manufacturing I'm able to work with 3D-printing, laser cutting and a wide variety of manual tools as well.

Internal Structures

Because of the internal transmission line structure and the way that sound must be bounced around the inside of the enclosure it must be possible to create quite specific internal shapes, this would be best done by 3D printing since any internal structure can be printed in place and no complex fixings are needed.

Precise Part Attachments

The best methods for precision (which is relevant for ensuring a good fit on the speaker, PCB mount and any other preexisting components) are 3D-printing and laser cutting. With laser cutting being potentially slightly better since parts will fail far less frequently.

Primary Techniques

Looking at all the information on this slide it seems the best method to use is to primarily 3D-print the speakers. This is because PLA is the densest material I have access to, and it means that I can design almost any shapes for the interior of the towers and that they'll be able to be manufactured with relative ease.

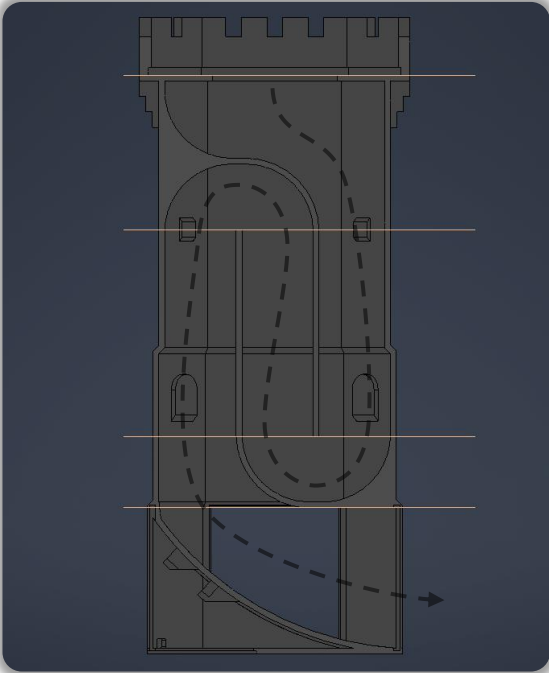
I may well use other methods as and when I need to, but I'll design the towers to be 3D-printed.

It is likely that I won't be able to print the entire tower in place because it's too big to fit on the print bed. And it would also be very difficult to assemble because I wouldn't be able to access the interior.

Fixings

For this design I'm going to use a screw over glue approach. This means that wherever possible I will join parts together with nuts and bolts, or with interference fit joints. This will make the speakers possible to disassemble so any maintenance that is needed can be done.

Interior Design

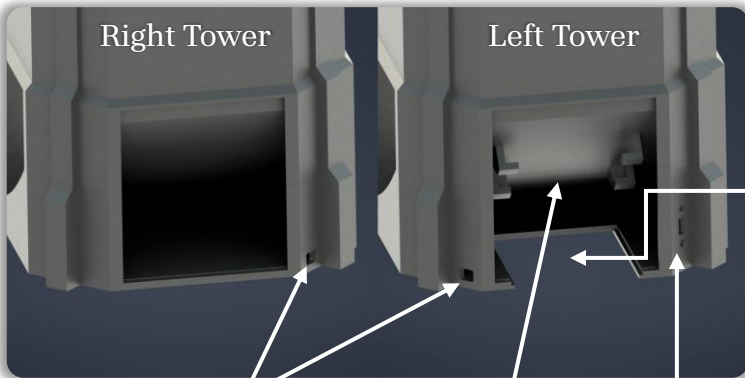
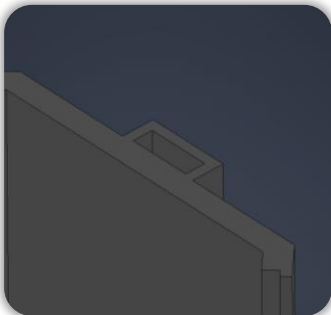
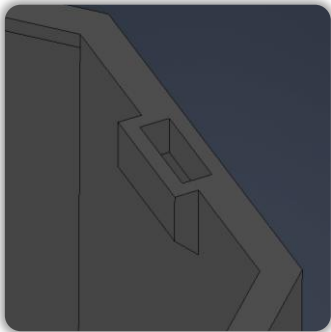


Transmission Line

The dashed line above shows the path the sound will take through the interior structure of the tower. The sound will still come out of the top as well, but anything out of the back of the speaker will travel through the transmission line and come out of the bottom of the tower. (Through the three holes in the bottom, see right). These holes will need covering with something, probably a laser cut grill or something similar since they don't look that good currently. I'll do the same to protect the speaker.

Divisions

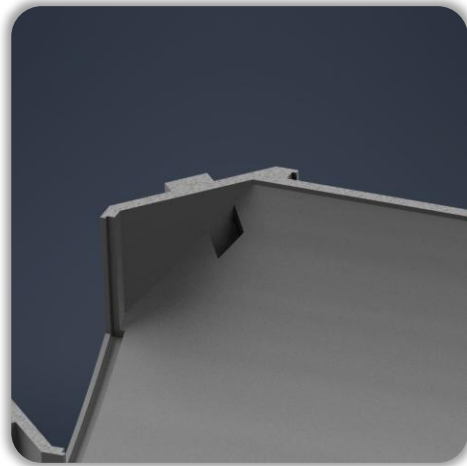
The orange lines are planes used to split the tower into 5 different parts. They are placed to try reduce the need for supports and to improve print stability. To join them I will use a small interference fit joint at each corner of the tower. The images on the right shows this clearly. Because of the electronics, the left and right towers must be subtly different from each other. To do this I have made use of inventors I-parts feature to enable or disable each feature as needed. You can see this in the bottom right images. This also allows me to easily generate the split parts to be printed.



Wire hole

Battery
mount

Switch
mount



Speaker Protection

There is a lip under the castellations. This is to allow a laser cut grill to be placed above the speaker to protect it from getting accidentally pierced.

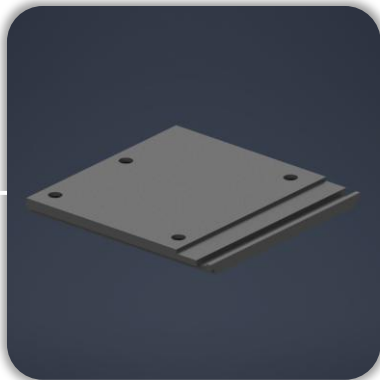
Wire

The hole on the left is to allow the wire to pass from where the PCB is up to the speaker. It will follow the path of the transmission line.



PCB Mount

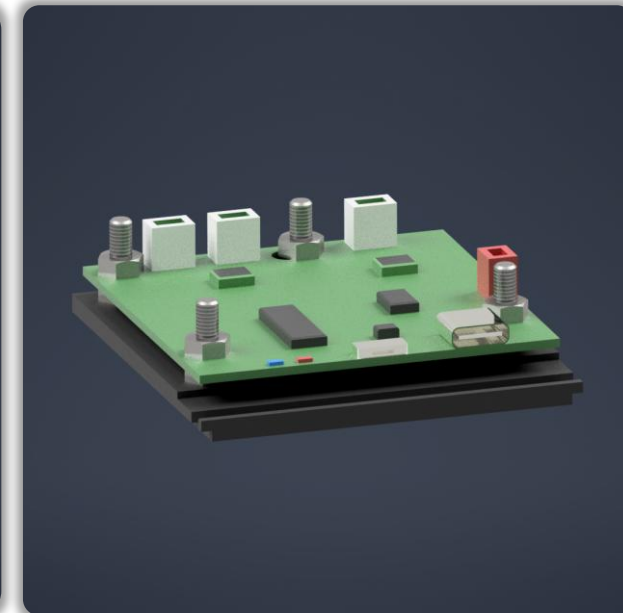
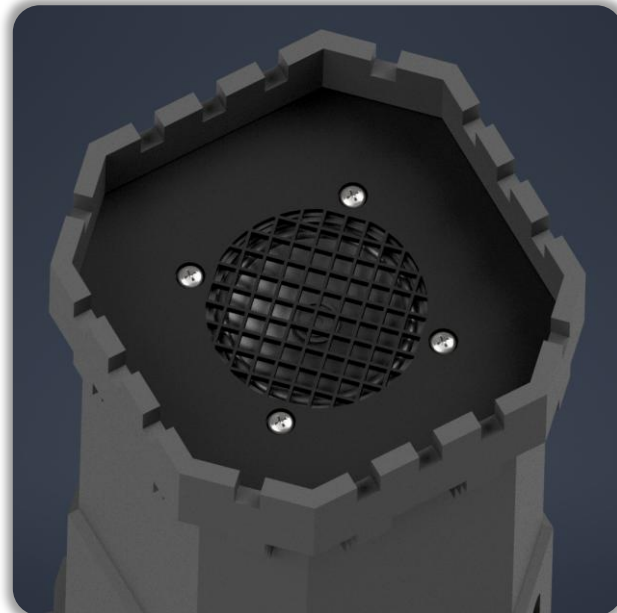
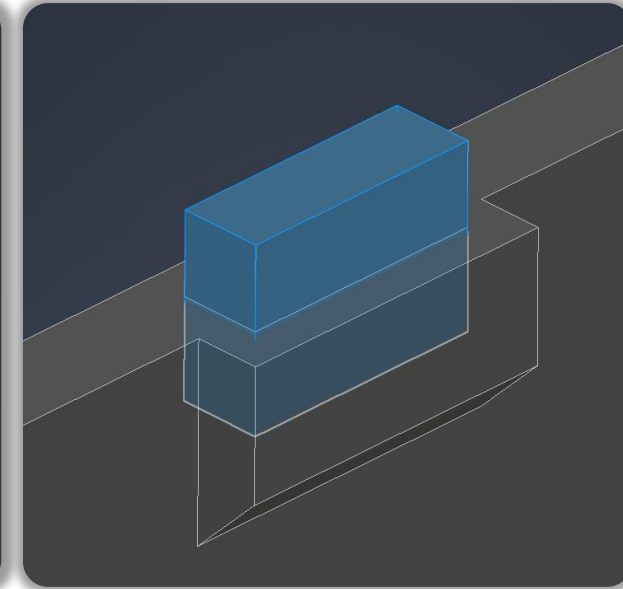
This is the mount for the PCB. It gets inserted into the base of the left tower.



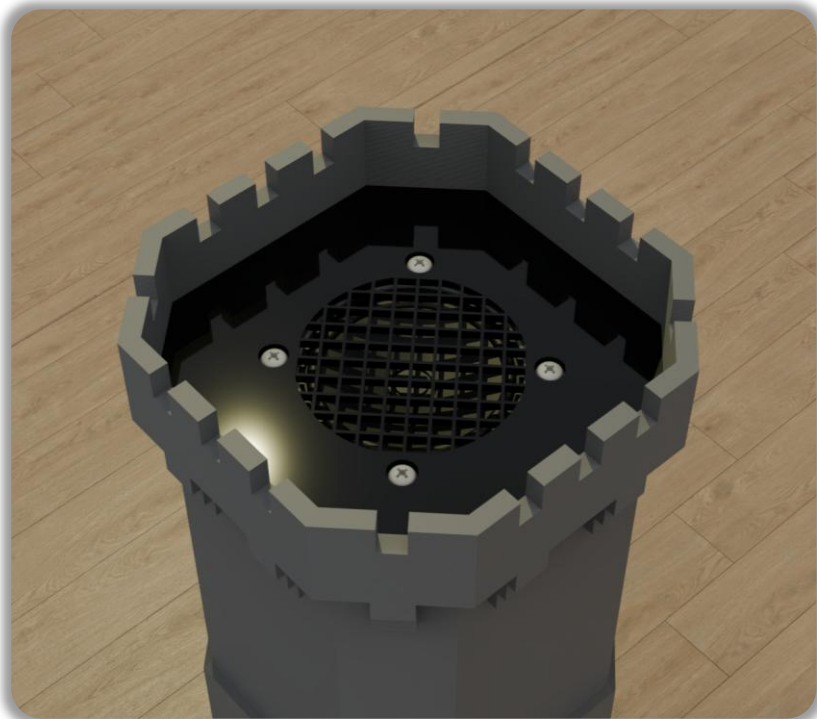
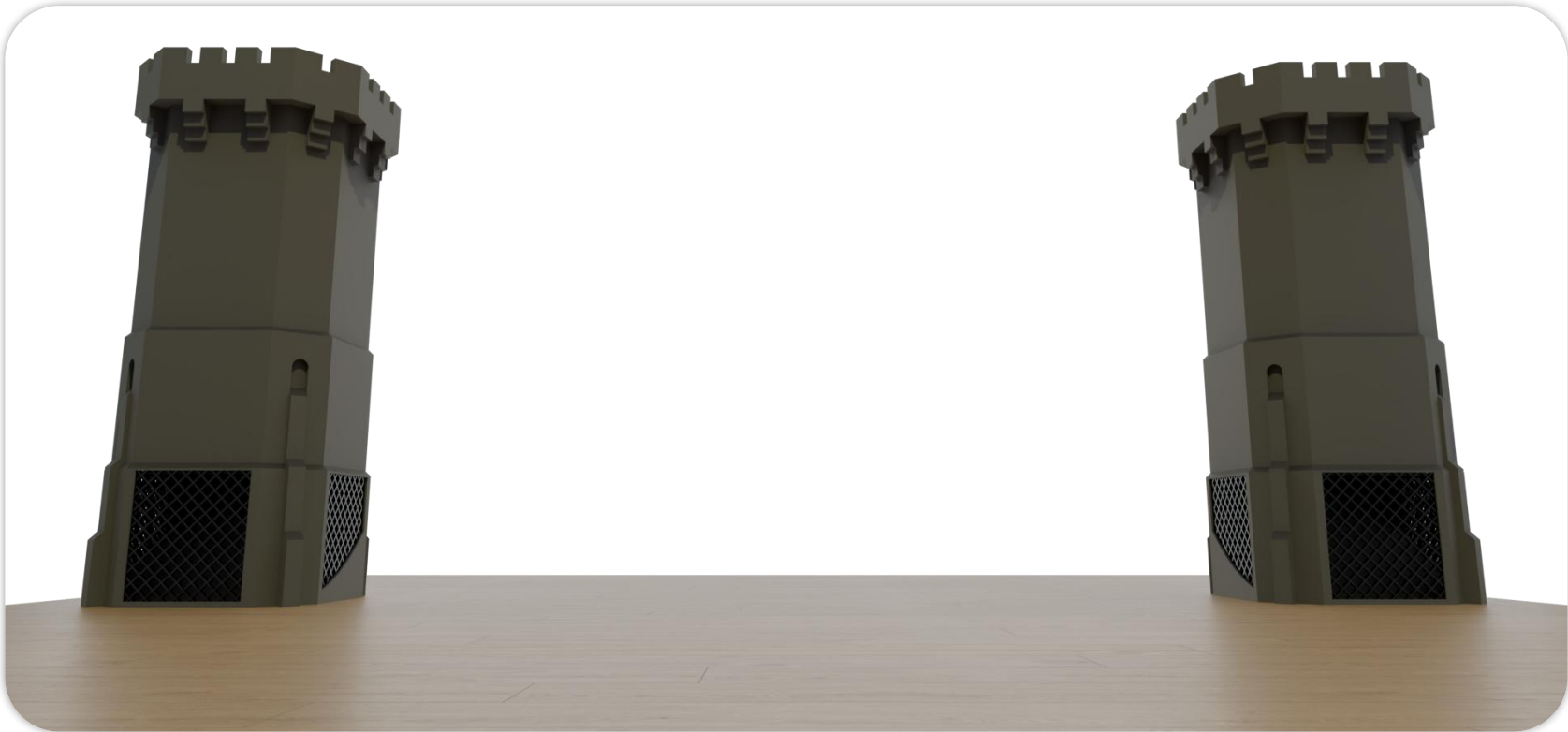
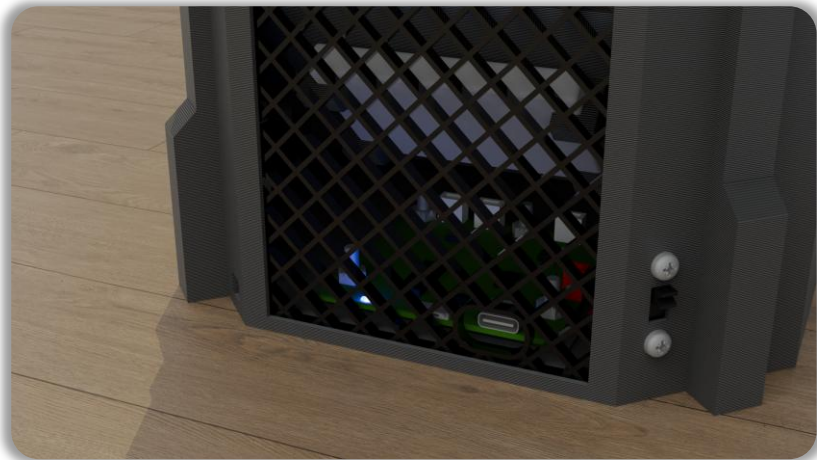
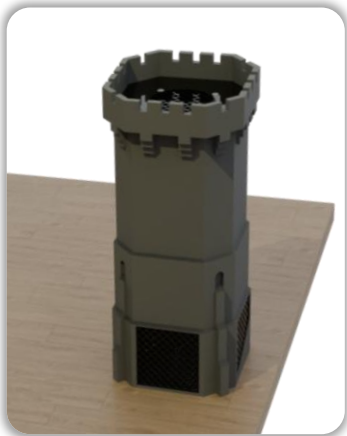
Digital Assembly



These images show features of the assembly I have made for the towers. The grills have been designed to look aesthetically pleasing while still allowing sound through. The one on the back of the left tower has a hole for the charging cable to pass through and allows light from the PCB to be seen. I also modelled the table they'd be placed on If used for a D&D session to gauge how they would look/feel in the environment that they'll be used in. The PCB has a layers of nuts to raise it above the mount to protect the underside from scratching. I used standard components from Inventors content libraries to allow me to generate a BOM later



Assembly Renders



BOM and Manufacturing Plan

The various tables around this page each have a function to help with planning of the manufacturing and assembly of the towers. I also decided to add an NFC tag (near field communication tag) to the design. I can program it to allows phones with the NFC software installed to connect to Bluetooth more quickly and easily.

PrintFile	Duration	QTY	Notes
TowerLeftPt0...	14:04:00	1	
TowerLeftPt1...	06:58:00	2	Same LR
TowerLeftPt2...	14:42:00	2	Same LR
TowerLeftPt3...	16:49:00	2	Same LR
TowerLeftPt4...	03:07:00	2	Same LR
TowerRightPt0...	13:51:00	1	
Fixing...	00:20:00	2	16 On plate + Spare
PCBMount...	00:36:00	1	

3D-Printing Plan

The table above shows the durations for each print file. The files don't match the BOM exactly because some of the tower parts are the same on the left and right towers. I used the Print durations to create the diagram below. Each square is an hour. The red and green shows the times that I can and can't set a new print running respectively and white is times that it might be possible for someone else to set one for me. This allowed me to create an efficient manufacturing schedule. The short prints aren't included since they are so quick they could be done anytime.

Part Number	Thumbnail	QTY	Manufacturing	Notes
Battery		1	N/A	with Wire and switch
Fixing-01		32	3D-print	+ Spare
GrillBack		2	Laser cut	
GrillFront		2	Laser cut	
GrillSide		4	Laser cut	
GrillTop		2	Laser cut	
NFCtag		1	N/A	Programmed for Bluetooth Quick Connect
PCB		1	N/A	
PCBMount		1	3D-print	
Speaker		2	N/A	One with 600mm Wire One with 1550mm Wire
Switch		1	N/A	Attached to battery Wire
TowerLeftPt0		1	3D-print	
TowerLeftPt1		2	3D-print	Same LR
TowerLeftPt2		2	3D-print	Same LR
TowerLeftPt3		2	3D-print	Same LR
TowerLeftPt4		2	3D-print	Same LR
TowerRightPt0		1	3D-print	NCF same as left
Bolt M2.5 x 8		2	N/A	No M2.5 Nuts, Switch held by these only
Bolt M3 x 12		4	N/A	For PCB mount
Bolt M4 x 12		8	N/A	For speakers
Nut M3		8	N/A	For PCB mount
Nut M4		8	N/A	For speakers

BOM

This is my final BOM for manufacturing. It lists the name of the part, an image of each one, the quantity, manufacturing method and any other notes that may be useful. As I manufacture, I can compare what I've got with this list to ensure that I have every component I need.

Part Number	Thumbnail	QTY	Manufacturing	Notes
Battery		1	N/A	with Wire and switch
Fixing-01		32	3D-print	+ Spare
GrillBack		2	Laser cut	
GrillFront		2	Laser cut	
GrillSide		4	Laser cut	

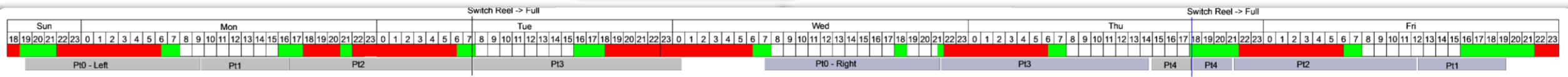


Filament Reels

The slicer I used to create the GCODES to print shows a predication for how much filament would be used. To ensure the filament was used efficiently I had to plan when I would changeover the reels. The tables below show the predicted weight used for each print file and a sum of the weights on each reel. The changeovers were then marked in the printing plan at the bottom of the page. I haven't maximised the usage on every reel, but I have ensured that none of them will run out of filament which is more important since the reels can be used for other prints

Part	Weight
Fixing	2.76
Mount	14.59
PT0Left	231.4
PT1	150.28
PT2	287.94
PT3	341.32
PT4	55.18
PT0Right	237.79

Reel 1		Reel 2		Reel 3	
Fixing	2.76	PT3	341.32	PT4	55.18
Mount	14.59	PT0Right	231.4	PT2	287.94
PT0Left	231.4	Pt3	341.32	PT1	150.28
PT1	150.28	PT4	55.18	Fixing	2.76
PT2	287.94		0		0
	0		0		0
	0		0		0
	0		0		0
Total	686.97	Total	969.22	Total	496.16



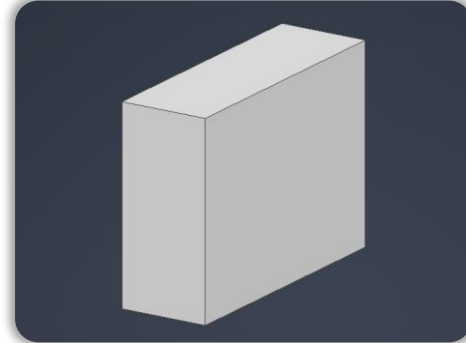
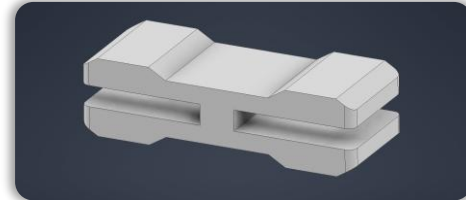
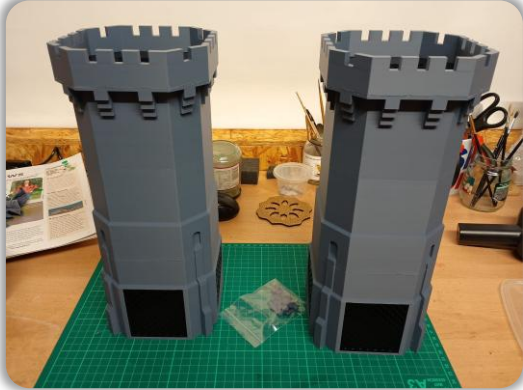


Manufacturing

3D-Printing

The 3D-printing went nearly perfectly, none of them failed and they all finished on-time or slightly earlier. I had to reprint the castellations part for each tower because the grill wouldn't fit under the lip at the top. All the parts had brim on them to ensure they stayed stuck to the print-bed, so this had to be cut off with a scalpel. Additionally, under the curved overhangs some of the lines of filament hadn't quite stuck so had to be either plucked off or glued down depending on how loose they were.

Additionally, the filament changeovers worked exactly as planned. Although in hindsight a single larger reel would have been better because no changeovers would have been needed, and this would have been more cost effective too. But I have made it work with what I had.



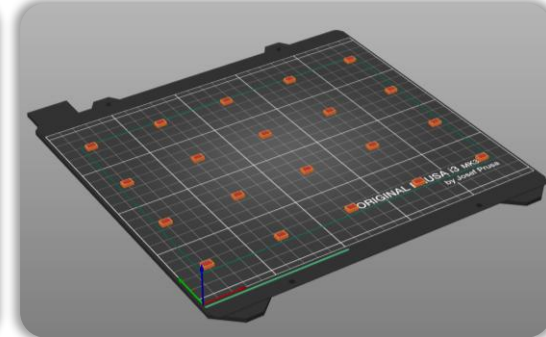
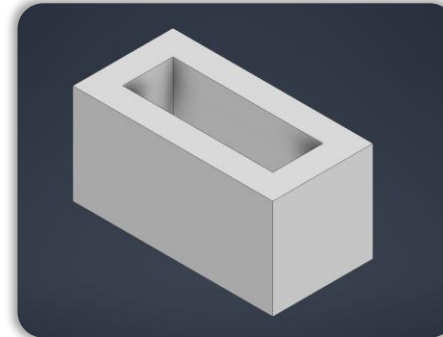
Laser Cutting

To cut the grills out I found some 3mm Black acrylic and used the CAD I'd already created to generate a DXF that could be sent to the laser cutter. Because of how small the parts are, there was slight warping of the pieces. For most, this was not a problem, but I had to adjust the design for the top grill because it was warping a lot. (the final design is the one pictured throughout this project) I also think that the newer design looks better overall.

Fixings

I Tried two different designs for the fixing, The first was slightly bigger than the hole so the gap in the middle allowed it to bend and hold against it. This was an impossible shape to print at the size I needed it. The other Was a simple cuboid that I thought might just provide enough friction to hold everything together.

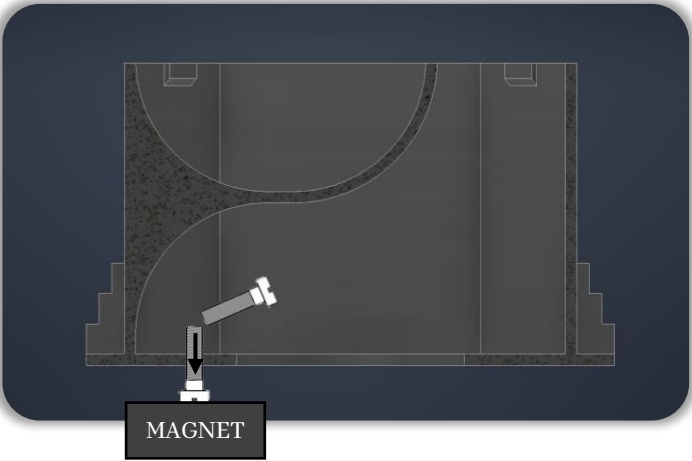
To work out what size of fixing was best I printed out several subtly different sizes and a 2 small pieces with a hole the size of the ones in the tower parts. I found that the best fit was to have fixings just slightly bigger than the holes they were going into. This allows them to rub slightly and stay tight. These were very prone to failing during printing due to their small size, but I didn't want to add brim because it would have added a lot of extra time and effort for relatively little benefit.



Wires

To allow the speakers to be placed where they need to in the towers, I needed to extend the wires and place a switch in one of the wires for the battery. One pair of wires needed to be at least 650mm and the other 1550mm. This was quite useful because it meant I could also strengthen the joins between the wire and the speaker. (Which were very weak initially)

Final Assembly



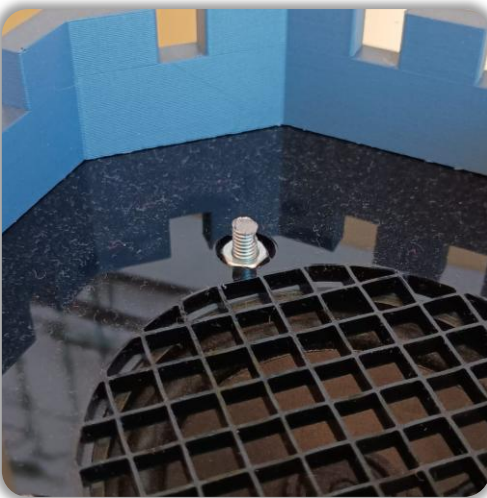
Attaching the speakers

Because of the shape of the part that the speakers attach to I had to deviate from the CAD slightly. I couldn't fit my hand it to screw the nuts in so instead I reversed the bolts, meaning the head was on the inside and the nut was on the outside. It was still difficult to find the holes, so I used a magnet to guide the bolts through. This also means it's easier to remove the speakers if something needs repairing. I had to shorten the bolts because of this, and since that meant the messy end of the bolts is visible, I had to design a part to cover them.



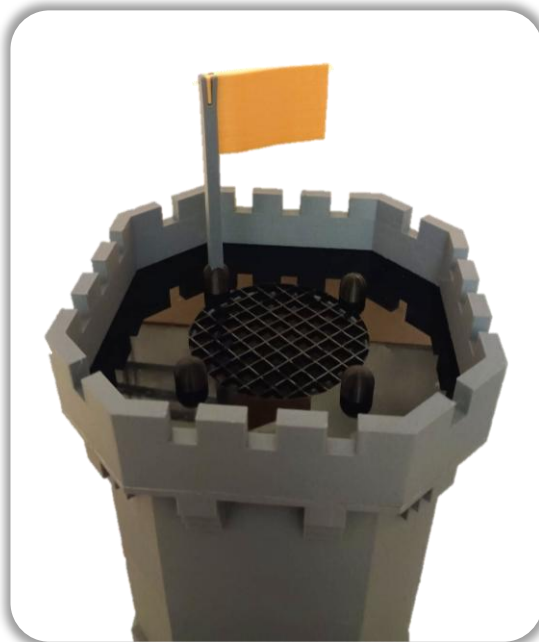
Wires

To improves the negative aesthetics of the wires that are joining the two towers I stripped a VGA cable and wrapped the outer cladding around them. It isn't a perfect join, but it does make it look much better



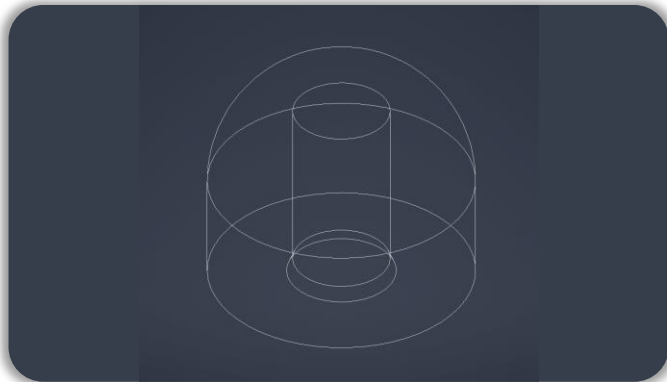
Joining the pieces

As planned, the fixing were tight enough to hold everything together. The only slight difficulty I had was with the castellations where the holes for the fixing was slightly smaller than designed (because of the width of the part the hole was in), but it just meant it was a slightly tighter fit. Although a couple of the grill pieces needed a little sanding, they ended up fitting perfectly. Finally, attaching the PCB to its mount and slotting that into the towers (along with the battery and the switch) worked exactly as planned and was loose enough to be easy to remove if needed.



The Flag

To add some interest to the top of the tower I decided to make a flag to cover one of the bolts. It was split into 3 parts to allow the base and flag to be different colours, and to remove the need for supports.



Storage Box

I also designed a box to store the speakers when not being used, it was padded to protect them and means that they can be transported more easily and safely.



Against the Brief

Original in Appearance

Use of the tower/castle aesthetic seems to be unique to my design. Even by googling explicitly for similar ideas I can't find a single one. Admittedly, this may be because there is a brand of speakers called "castle", and a tower is a type of speaker so results for something that actually looks like a castle or tower may get hidden. Regardless I believe this means that when it comes to originality I have fully succeeded.



Innovative use of Materials

The combined use of laser cut and 3D-printed parts is not unique to this product nor is the use of removable and interference fit fixings, but throughout I have tried to keep good practices and to allow disassembly. And in these regards I have fully succeeded. I probably wouldn't use the word innovative to describe my use of materials, but I would say that I have used them in ways that are very effective for this project.

Aimed at a Specific Consumer Market

By using quite a fantasy theme, and because of the focus on making the speakers not look like they are speakers. I feel that I have appealed quite well to the DND/TTRPG market because it means the speakers can be used as props on the game table but also be functional as speakers to provide ambiance, sound effect and/or music. This means I have succeeded at creating a product for a specific customer market.



Against my Requirements

Sound Quality

I am very happy with the sound quality of the final speakers. There are probably various areas for improvement and with more knowledge of the physics of sound waves and of speakers I could probably work out how to do those improvements. But given how much I don't know, I feel I've managed to done a good job of creating a decently bass-y but clean sound. With effective use of materials, I've created a heavy object to try avoid wasted energy. The transmission line structure defiantly plays a large role in the sound quality alongside the avoidance right angles and attempt to create curves to guide sound where I want it. Overall, I'm really happy with the sound quality



Maintenance

I wanted the speakers to be easily maintainable in case something breaks. I have achieved this by making all of the joints with fixings that can be undone and parts that can be separated. The PCB can be removed with its mount and all wires can be unplugged so if any of the wires break, they can be repaired. The only real negative is the speaker cones because although they can be removed, they are very difficult to re-fix again. If one of the 3D-printed parts breaks, it would have to be re-printed, but because they can all be separated, it would be possible to replace just one. The laser cut parts are quite delicate but again, can be easily replaced. For all these reasons I'm really pleased with the ability to maintain the speakers.

